

Embedded Components

Design Guide

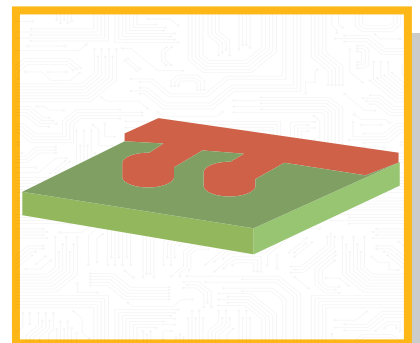
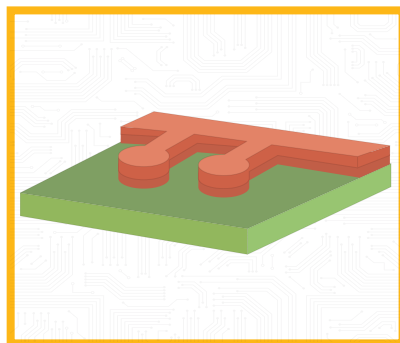
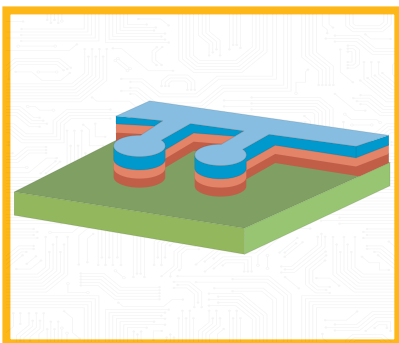
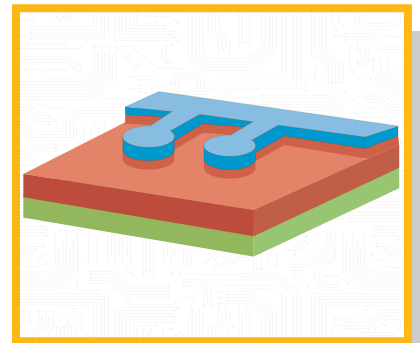
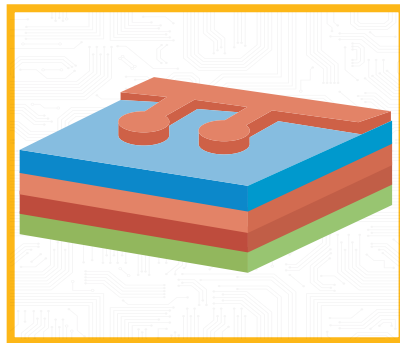
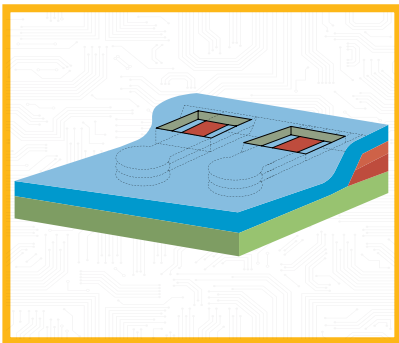


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1. Introduction

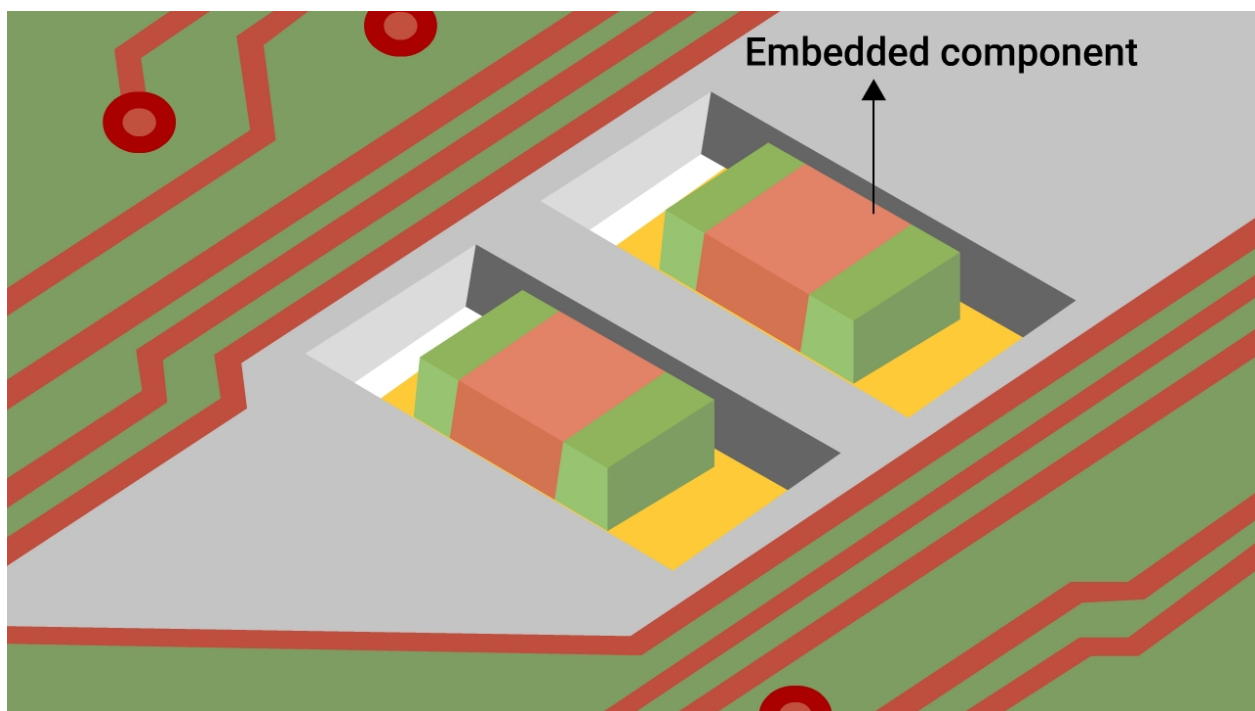
Embedded components are typically integrated directly into the internal layers of a PCB during fabrication. This technology eliminates the need for SMT components, enabling significant reductions in the board form factor.

By leveraging embedded passives, you can enhance signal integrity, reduce parasitic effects, improve power distribution, and achieve superior high-frequency performance. This technique is advantageous for compact, high-density applications, including smartphones, wearable devices, implantable medical electronics, aerospace systems, and next-generation computing platforms.

This design guide provides the best practices for successfully incorporating embedded passives into your PCB designs. It also covers key benefits, material and fabrication considerations, layout, and simulation strategies.

1.1 What are embedded components?

Embedded components are passive electronic parts, such as resistors and capacitors, that are integrated within the internal layers of a printed circuit board.



An illustration of embedded passives integrated into a PCB substrate

Embedded components are classified into:

- **Printed passive elements:** Formed by depositing conductive or dielectric materials directly onto inner copper or dielectric layers.
- **Thin/thick-film components:** Used to create high-precision resistors or capacitors by depositing materials through photolithographic or screen-printing methods.

- **Discrete embedded parts:** Placed in cavities or between layers before lamination, allowing for compact and integrated circuit designs.

You can opt for integrated components if you're working on:

- High-density designs, where form factor is crucial.
- High-speed and RF circuits, minimizing parasitic effects for superior signal integrity.
- Critical systems, including automotive, aerospace, and medical devices, where reliability is paramount.

1.2 Benefits of using embedded resistors and capacitors

1.2.1 Reduced form factor

By integrating components within the PCB layers, you can significantly reduce the board's surface area requirements. This reduces the need for surface real estate typically occupied by SMDs, enabling smaller, lighter, and more compact devices.

For example, in mobile devices, embedded capacitors can free up space for other critical components like antennas and sensors.

1.2.2 Lower parasitic effects

Embedded components are generally positioned close to the circuit's active elements, reducing the length of interconnects (traces and vias). Shorter interconnects minimize parasitic inductance and capacitance. This leads to cleaner signal paths, minimizing the risk of signal interference and reflections.

For instance, embedded capacitors in power distribution networks (PDNs) help decouple high-frequency noise more effectively.

1.2.3 Improved high-frequency behavior

At high frequencies, traditional SMDs experience parasitic losses due to solder joints and long traces.

Embedded parts, with their direct integration and shorter connections, reduce impedance discontinuities and insertion losses. This ensures stable signal propagation and reduces loop inductance in RF circuits.

2. SMD vs. embedded components

2.1 Design trade-offs: size, performance, cost

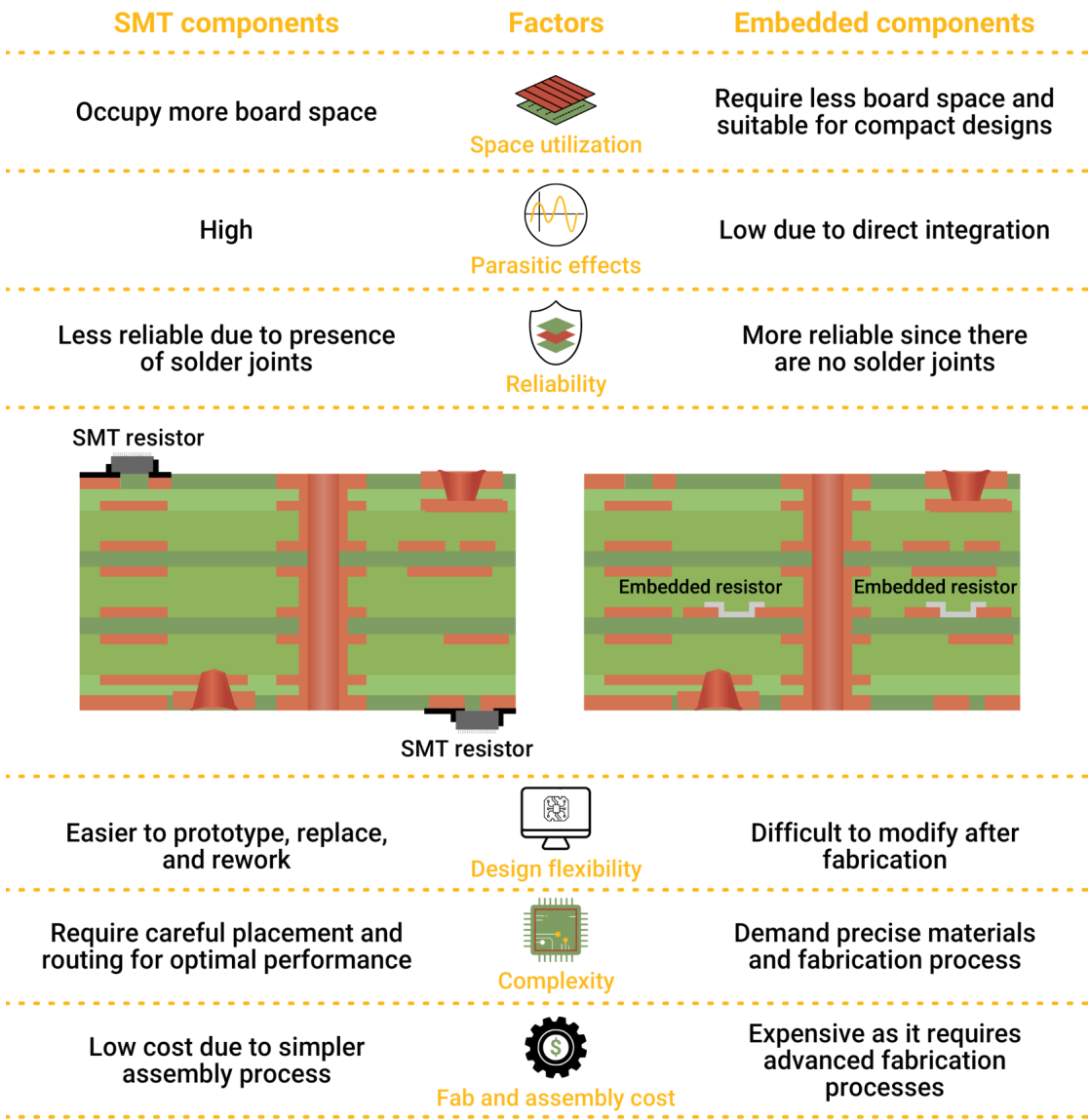
When deciding between SMDs and embedded components, you must consider trade-offs in board size, electrical performance, and overall cost.

The table below breaks down these tradeoffs:

Parameter	Embedded components	SMD components
Size	Save surface space, enabling denser layouts and smaller PCB footprints.	Available in very small packages (e.g., 0201, 01005) but still occupy board surface and require clearance for placement and soldering.
Electrical performance	Reduce parasitic effects and enhance signal integrity due to proximity and integration.	Offer a wider range of values and tighter tolerances, essential for precise circuit tuning.
Production costs	Increase fabrication costs due to specialized materials and processes (e.g., laser trimming, resistive/capacitive laminates).	Generally lower component and sourcing costs, but may increase assembly costs due to pick-and-place and soldering steps.

The infographic below summarizes the difference between embedded and SMT components:

SMT Vs. Embedded Components



2.2 When to choose embedded over SMT

The choice between embedded parts and SMDs depends on the form factor, performance, and budget of your design.

Embedded components are preferred when:

1. Space constraints are critical

In ultra-compact designs like wearable electronics, IoT modules, implantable medical devices, and smart cards, PCB real estate is limited.

Laminated components free up valuable surface area and reduce the size of the end product.

2. High-frequency performance is required

Applications such as RF transceivers, 5G modules, high-speed memory interfaces, and mixed-signal circuits benefit significantly from embedded passives. They reduce the length of interconnections and eliminate parasitic inductance and capacitance associated with SMT pads and vias.

This results in:

- Better signal integrity.
- Lower EMI.
- Improved power delivery.

3. Applications demand high reliability

Integrated parts eliminate solder joints, which are often a failure point in environments with vibration, thermal cycling, or mechanical stress.

Incorporating embedded components reduces these risks, making them ideal for aerospace/defense, automotive, and industrial applications.

In many practical designs, a hybrid approach combining embedded passives with surface-mount technology offers the best of both worlds. While embedded passives are ideal for saving space and enhancing electrical performance, they may not always meet every design requirement. Some component values, tolerances, or power ratings might not be readily available in the embedded form.

For example, low-value decoupling capacitors can be embedded to reduce loop inductance and save board space. In contrast, high-value bulk capacitors can be placed on the surface to meet specific electrical needs.

Similarly, precision resistors often remain as SMD due to tighter tolerances or the need for future rework.

Collaborate with your contract manufacturer to find the best way to use embedded and surface-mount parts.

3. Key factors to consider when selecting integrated passives

Selecting embedded resistors and capacitors requires a thorough understanding of their electrical, thermal, and mechanical properties.

Here, you'll learn the key factors to consider when picking embedded passives for your designs:

3.1 Choosing embedded resistors

Embedded resistors serve various functions such as termination, biasing, and voltage division. Their effectiveness depends on target resistance range, tolerances, power ratings, and temperature coefficients.

3.1.1 Target resistance range

The resistance range is determined by the material's resistivity and how it is processed

The table below shows the target resistance range for different types of integrated resistors:

Resistor type	Typical resistance range	Characteristics	Use cases
Thin-film resistors	10Ω to 100kΩ	Low noise and low temperature coefficients	Analog, RF, and high-speed digital applications
Thick-film resistors	up to 10MΩ	Less precise than thin-film and cost-effective	General-purpose electronics
Printed resistors	1 Ω to 10MΩ	Resistance values are less predictable due to variations in ink thickness and curing	Low-precision designs

Always verify that your required resistance range is compatible with the fabrication process and the type of material used.

3.1.2 Tolerances

Tolerance defines how much the actual resistance can vary from its nominal value. Choosing appropriate tolerance levels is crucial for maintaining consistent impedance.

For applications like analog filters and oscillators, use resistors with tighter tolerances (e.g., ±0.1%) to ensure precision and stability.

However, achieving tight tolerances may require additional fabrication steps such as laser trimming, which can increase manufacturing costs. You must carefully balance the need for precision with overall cost constraints.

To calculate the dimensions of an embedded resistor, use the formula:

$$R = (\rho \times L) / (w \times T)$$

Where,

R: desired resistance value (Ω)

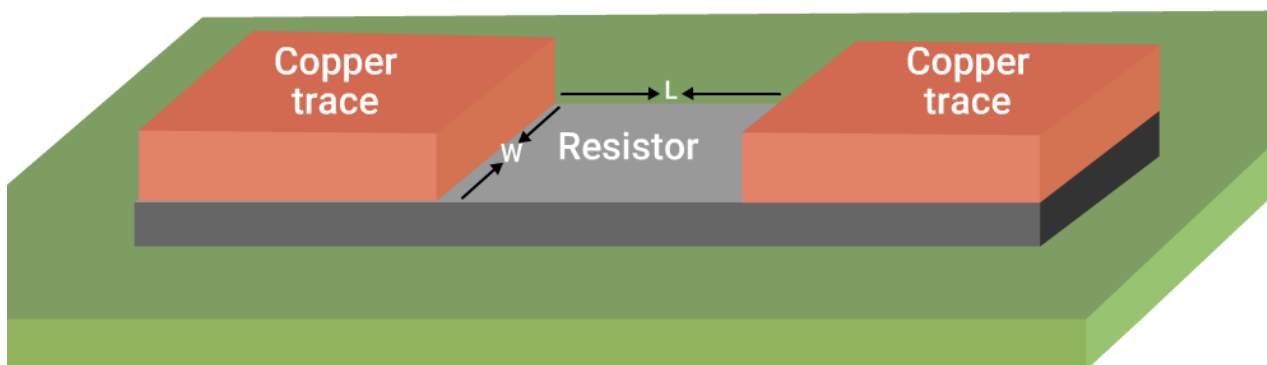
ρ : resistivity of the resistive material ($\Omega \cdot m$)

L: length of the resistor (m)

W: width of the resistor (m)

T: thickness of the resistive layer (m)

Adjust the length/width to modify the resistance. Ensure dimensions meet fab tolerances and fit within the available PCB space.



Modify the embedded resistor length/width to achieve the desired resistance

3.1.3 Power ratings

Power handling capacity of embedded resistors is inherently limited due to restricted heat dissipation. Power rating mainly depends on:

- Resistive film's thermal conductivity.
- Surrounding copper area.
- Board thickness.

To ensure safe operation, evaluate the resistor's maximum current-carrying capacity based on its power rating and resistance value. This is especially important for resistors formed using thin and narrow traces, as they tend to have lower current-carrying capabilities.

When finalizing embedded resistors, always ensure that the power rating is high enough to prevent overheating. Inadequate ratings can lead to issues such as local delamination, hot spots, or thermal drift in resistance over time.

Typically, embedded resistors are rated between 50 mW and 250 mW.

3.1.4 Temperature coefficient of resistance

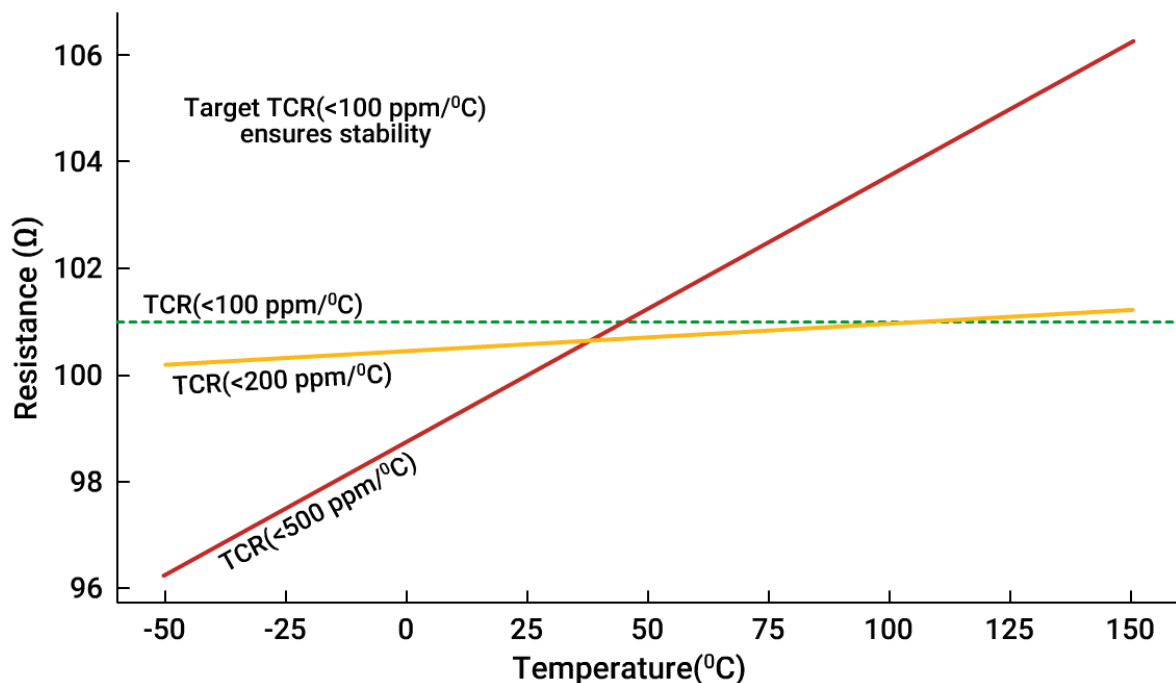
The temperature coefficient of resistance quantifies how resistance changes with temperature variations. It is expressed in parts per million per degree Celsius ($\text{ppm}/^{\circ}\text{C}$). It indicates the relative change in resistance per degree of temperature change.

Always select resistors with a low temperature coefficient of resistance to minimize resistance variations due to temperature fluctuations.

A low temperature coefficient of resistance is essential to:

- Prevent significant resistance changes that could affect circuit performance, especially in analog or precision digital circuits.
- Ensure consistent behavior across a wide temperature range, which is critical for applications in varying environmental conditions.

For optimal performance, aim for values below $100 \text{ ppm}/^{\circ}\text{C}$. This ensures stability even with temperature fluctuations (e.g., -40°C to 85°C , common in industrial electronics).



Resistance vs. temperature for resistors with different temperature coefficients

In the graph above, the red line represents a resistor with a high TCR (500 ppm/°C), showing a significant slope, while the yellow line represents a resistor with a TCR (200 ppm/°C), showing a gentler slope. The green line highlights the target TCR range (<100 ppm/°C) for stable circuit performance.

3.2 Choosing embedded capacitors

Embedded capacitors are commonly used for decoupling, bypassing, and filtering applications. Their effectiveness depends on capacitance range, tolerance, voltage rating, and frequency response characteristics like ESR and ESL.

3.2.1 Capacitance range

Embedded capacitors typically range from 1 pF to 100 nF, with some specialized materials offering higher values. The capacitance depends on the dielectric material, layer thickness, and electrode area. For example, high D_k laminates can achieve higher capacitance per unit area.

3.2.2 Tolerances

Integrated capacitor tolerances are generally wider than SMD capacitors, ranging from $\pm 10\%$ to $\pm 15\%$. Applications requiring precise capacitance values may need hybrid designs with embedded and SMDs.

3.2.3 Voltage rating

Voltage ratings for these capacitors typically range between 5 and 100 V. You must ensure the voltage rating exceeds the maximum operating voltage of your design to prevent breakdown.

3.2.4 ESR and ESL

ESR (equivalent series resistance) and ESL (equivalent series inductance) are critical for high-frequency performance. Embedded capacitors generally have lower ESL due to their proximity to the circuit. ESR depends on the dielectric material. Low-ESR materials like ceramic-based laminates are preferred for decoupling applications.

4. Types of embedded resistors

Embedded resistors can be manufactured using several techniques, each offering trade-offs in precision, cost, processing complexity, and performance.

The choice of resistor type depends on your application's requirements, such as tight tolerance for analog circuits and cost-effective solutions for consumer electronics.

Here, you'll learn the four most common embedded resistor technologies, printed, plated, thin-film, and thick-film, along with their materials, advantages, limitations, and typical use cases.

4.1 Printed resistors

Printed resistors are made by applying a special resistive ink or paste directly onto the PCB. This can be done on a dielectric layer or inside a cavity. Common methods include screen printing or inkjet printing.

Materials:

Typically, carbon-based polymer thick films or conductive ceramic pastes are used.

Advantages:

- Low-cost and scalable for mass production.
- Compatible with additive manufacturing processes, allowing for simplified and flexible fabrication.

Limitations:

- Wide resistance tolerances (typically $\pm 15\%$).
- Poor stability over temperature and time.
- Limited performance at high frequencies due to parasitic effects.

Best used for:

- Low-power, non-critical functions such as pull-up/pull-down resistors.
- Consumer electronics, toys, and simple embedded control boards.

4.2 Plated resistors

Plated resistors are formed by electroplating a resistive alloy onto predefined copper patterns on a PCB.

Materials:

Nickel-phosphorus (NiP) alloy is electroplated onto copper traces.

Advantages:

- Improved tolerances ($\pm 10\%$ to $\pm 15\%$) compared to printed resistors.
- Better adhesion and uniformity than printed resistors.
- Resistance values can be fine-tuned post-deposition using laser trimming, allowing more control during fabrication.

Limitations:

- Suitable only for moderate-precision applications.
- Increased process complexity due to the need for precise plating control.
- Variability in plating thickness can affect resistance consistency.

Best used for:

- Embedded pull-up/down or termination resistors
- Medium-performance analog and digital circuits

4.3 Thin-film resistors

Thin-film resistors are fabricated by sputtering or vacuum-depositing a thin layer of resistive material onto a prepared substrate. The resistor pattern is then defined using photolithography and etching.

Materials:

Common materials include nickel-chromium (NiCr), tantalum nitride (TaN), and other metal alloys with low temperature coefficients and good environmental stability.

Advantages:

- High precision resistance with tight tolerances ($\pm 1\%$ to $\pm 5\%$, or even tighter after trimming).
- Excellent thermal and environmental stability.
- Minimal parasitics, ideal for high-frequency circuits.
- Resistance can be laser-trimmed after deposition for exact value matching.

Limitations:

- Higher fabrication cost due to complex processing steps.
- Requires cleanroom environments and advanced equipment.

Best used for:

- Aerospace, medical, and defense applications.
- Precision analog and RF designs.
- Circuits requiring tight resistance tolerances and long-term stability.

4.4 Thick-film resistors

Thick-film resistors are created by screen-printing a thicker layer of resistive paste, typically a ceramic-metal composite, onto the PCB substrate. The layer is then fired or cured to achieve the final resistive properties.

Materials:

Common materials include ruthenium oxide, silver-palladium, or other metal-oxide compounds suspended in a glassy matrix.

Advantages:

- Cost-effective and suitable for high-resistance values (in the kilo-ohm to mega-ohm range).
- Compatible with conventional screen-printing techniques used in PCB fabrication.
- Robust under moderate environmental conditions.

Limitations:

- Resistance tolerances are typically $\pm 10\%$ to $\pm 20\%$.
- Less suitable for high-frequency signals due to larger parasitic inductance and capacitance.
- Limited precision compared to thin-film alternatives.

Best used for:

- General-purpose signal conditioning, biasing, or pull-up applications.
- Automotive, industrial, or consumer-grade designs where cost is a key factor.

Comparison of different embedded resistor types:

Type	Typical tolerance	Materials	Applications
Printed resistors	$\pm 15\%$ to $\pm 20\%$	Carbon/polymer inks	Low-cost, non-critical applications
Plated resistors	$\pm 10\%$ to $\pm 15\%$	Nickel-phosphorus	Moderate-precision analog/digital circuits
Thin-film resistors	$\pm 1\%$ to $\pm 5\%$	NiCr, TaN	High-frequency circuits
Thick-film resistors	$\pm 10\%$ to $\pm 20\%$	Ruthenium oxide, Ag-Pd paste	General-purpose electronics

5. Material selection and stack-up design strategies

Designing for embedded passives requires careful material selection and thoughtful stack-up planning to ensure manufacturability and mechanical reliability. The following guidelines will help you optimize your embedded resistor and capacitor implementation.

5.1 Selecting the right laminate for resistive and capacitive layers

The performance of embedded resistors and capacitors is highly dependent on the materials used. Each embedded resistive or capacitive layer demands specific properties from the laminate.

Types of resistive layers include:

OhmegaPly: A nickel-phosphorus resistive foil laminated onto standard substrates. This is laminated to a dielectric material and subtractively processed to produce planar resistors. Due to its thin nature, it can be buried within layers without adding thickness. Here, epoxies, polyimides, and PTFEs are all acceptable organic substrates.

Advantages of OhmegaPly:

- Reduced noise, crosstalk, and EMI.
- Lower series inductance and shorter signal paths.
- Enhanced line impedance matching.
- Better wiring due to the elimination of via.
- Good reliability due to the elimination of solder joints.
- Reduced form factors.
- Improved active component density.
- Better assembly yield.
- Compatible with lead-free alloys.

Applications of OhmegaPly:

OhmegaPly materials are used in:

- Interposer board.
- Pull-up resistors in avionics.
- Series termination resistors in BGA packages.
- HDI PCBs.
- Telecom switching cards.
- OhmegaPly resistors in flex boards.
- Buried resistor applications.

TCR material: Thin-film controlled resistive foils available in a range of resistivities. TCR is an integrated thin-film resistor foil (grade 3 copper foil) that is available in different foil widths and thicknesses. The commonly available thickness values are 18 μm (0.5 oz) and 35 μm (1 oz).

TCR foil is available in three different resistivity alloys:

- Aluminum silicon (AlSi).
 - Chromium silicon monoxide (CrSiO).
 - Nickel chromium (NiCr).
- NiCr, when used with TCR, lowers fabrication steps by removing the conditions for a separate resistive layer etching.
 - Each of the three resistivity alloys can withstand multiple thermal excursions, like lead-free reflow, with minimal change to resistance and long-term stability.
 - The board processing method follows the same procedure as OhmegaPly. The only difference is that TCR laminate material is used instead of OhmegaPly laminate.

These resistive foils are typically bonded to standard FR-4 or high-performance laminates (e.g., polyimide, PTFE) to create embedded resistors.

When selecting a resistive material, ensure it has a stable temperature coefficient of resistance (TCR) and is compatible with subtractive etching and laser trimming processes.

Advantages of TCR

- Best PCB real estate utilization.
- Lead-free.
- Easily recyclable.
- Withstands multiple heat excursions.

In chapter 9, we will cover the step-by-step procedure of fabricating embedded resistors using OhmegaPly and TCR.

Types of capacitive layers include:

3M embedded capacitance material (ECM) and FaradFlex: These are commonly used ultra-thin laminates with high capacitance per unit area, suitable for embedded decoupling applications.

These materials are typically bonded between power and ground layers to create planar embedded capacitors within the PCB.

Always select a material with a stable D_k (3.5 – 4.5) and low dissipation factor D_f (<0.005) for high-frequency PCBs.

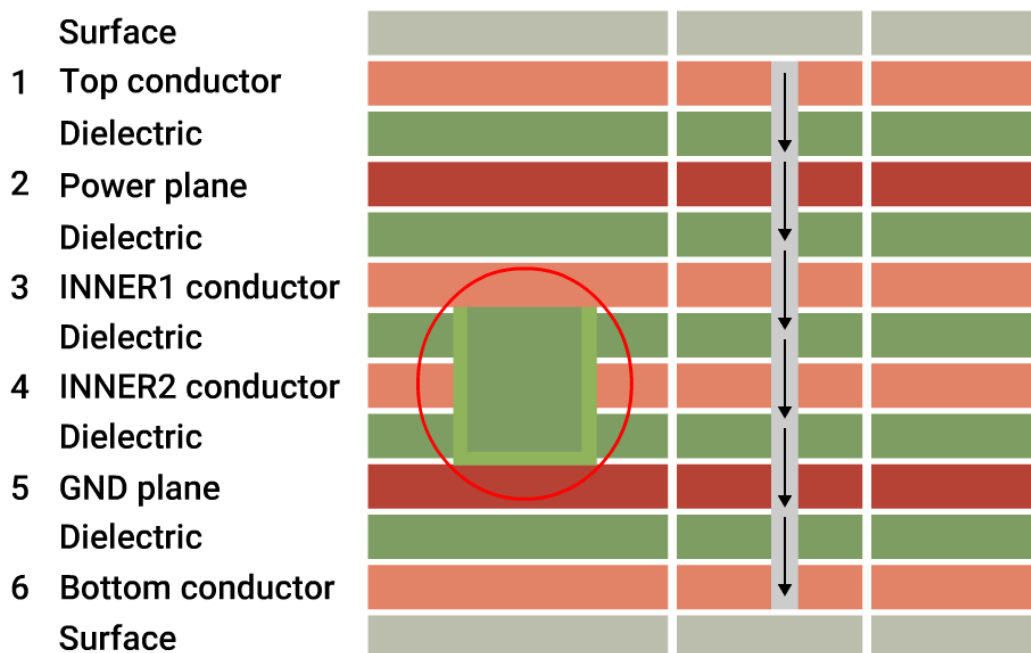


The graphic features a line graph on the left with 'COST' on the y-axis and 'PERFORMANCE' on the x-axis. A red line with a heart icon indicates a path of optimal material selection. To the right, the text 'PCB DESIGN TOOL' and 'Material Selector' is displayed in a bold, sans-serif font. Below this, there is a red calculator icon and the text 'TRY TOOL'.

5.2 Designing an efficient stack-up for embedded passives

Follow these guidelines when you're designing a build-up with embedded components:

- Place resistive layers near signal layers to reduce trace lengths and minimize parasitic inductance.
- Implement capacitive layers between power and ground planes to reduce loop inductance and improve decoupling.



An illustration of an embedded capacitor in a PCB stack-up (marked in red)

- Maintain a symmetrical stack-up to balance the mechanical stresses during lamination. For example, if a resistive layer is added on one side of the core, include a balancing layer (e.g., dummy copper plane or copper thieving) on the opposite side.
- Ensure sufficient clearance between embedded components and adjacent layers to avoid interference.
- Assign dedicated layers for resistive and capacitive elements to avoid routing issues. For example, use inner layers for embedded capacitors while reserving separate layers for resistors to simplify fabrication and testing.
- Isolate embedded components from high-speed signal traces to prevent crosstalk or electromagnetic interference. Use ground planes or via stitching to shield sensitive signals.



6. Design guidelines for integrating embedded passives in PCBs

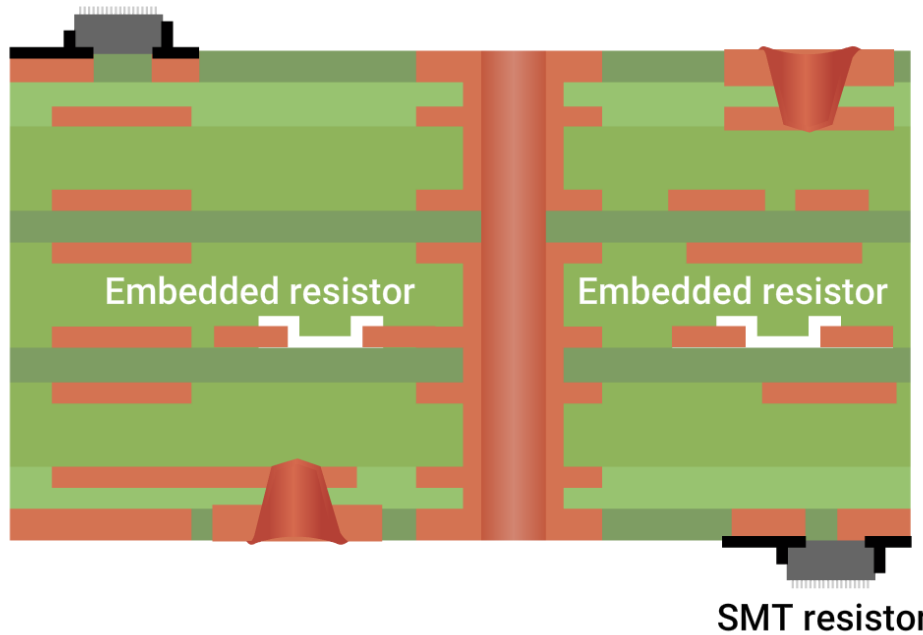
Integrating embedded passives into PCBs requires careful planning to ensure electrical performance and manufacturability. The following strategies focus on optimizing component placement and layer allocation to maximize the benefits of embedded resistors and capacitors.

6.1 Layout tips for integrating embedded components

Effective layout design is critical to leverage the advantages of embedded passives. Stick to these guidelines when you're designing your layout:

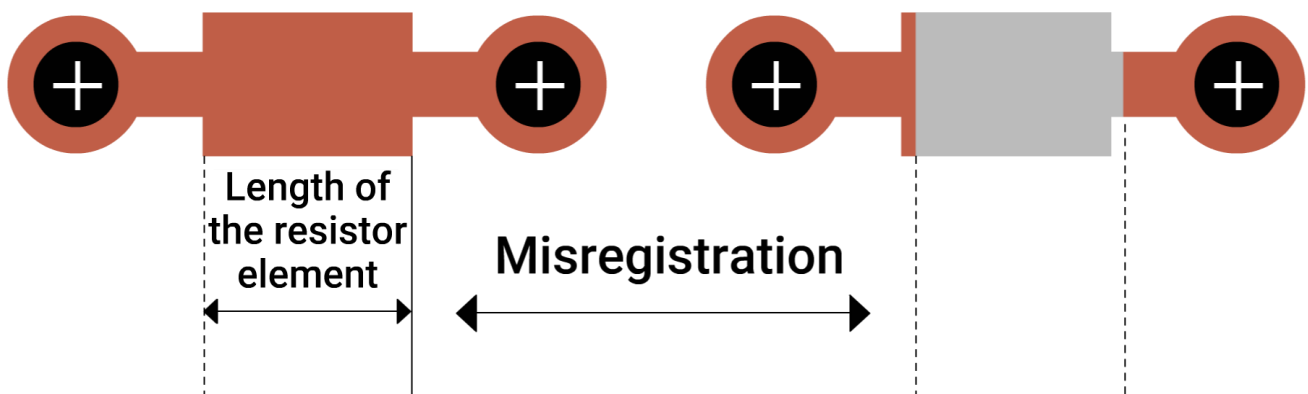
- Place embedded components as close as possible to the active devices (e.g., ICs or power management units) they serve. This minimizes trace lengths, reducing parasitic inductance and resistance. For example, embedding decoupling capacitors beneath a microprocessor can improve PDN performance.
- Position embedded capacitors between power and ground planes to shorten current loops, enhancing decoupling efficiency.
- Use thermal simulation tools to identify potential hotspots and avoid placing high-power embedded resistors in areas with poor heat dissipation.

SMT resistor



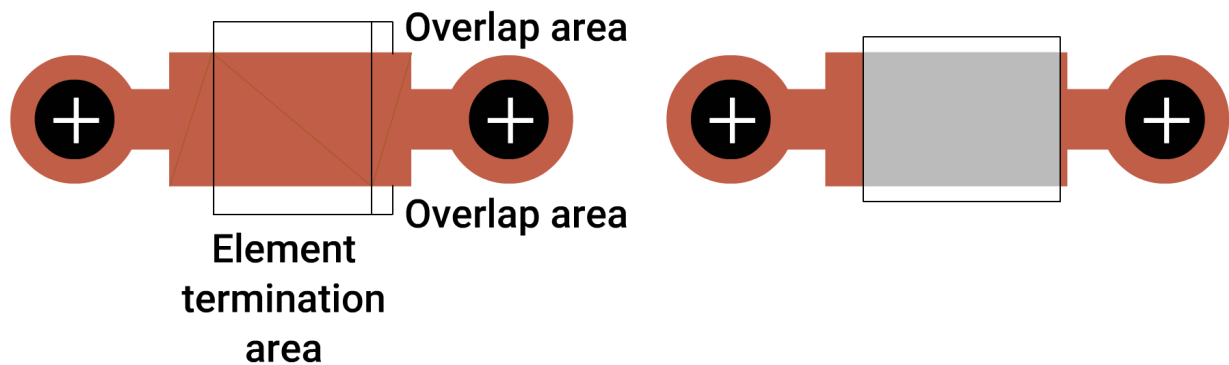
Embedded resistors offer better power dissipation capacity than SMT resistors

- Extend the length of the resistor element (5–10 mil) to compensate for any misregistration or misalignment of the artwork. The following image shows the composite's design elements during the first print.



Element misregistration during the first print

To balance any misregistration or misalignment of artwork in the image printing process, the overlap area should be 5 to 10 mil beyond the resistor's width. The image below gives an accurate representation of the overlap region.



The overlap area should be 5 to 10 mil beyond the resistor's width

- Distribute embedded components evenly across the board to maintain mechanical stability and prevent warpage during lamination. Uneven placement can stress the board, especially in multilayer designs.
- Run DRCs to verify parameters such as minimum spacing between embedded components, trace width, and spacing.
- Follow your manufacturer's design guidelines for embedded passives, as they may have specific requirements for layer thickness and material tolerances. For example, some manufacturers may limit the maximum size of embedded capacitors due to lamination constraints.

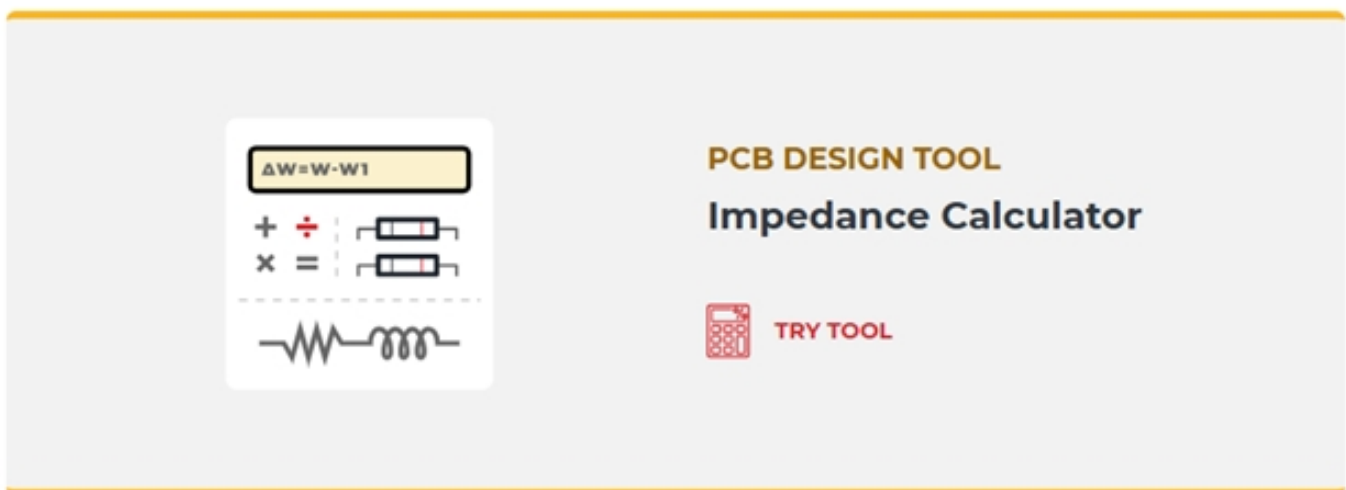
6.2 Optimizing trace and pad connections

The interconnects between embedded components and other circuit elements significantly impact overall performance.

Adhere to these guidelines when designing traces and pads:

- Keep the traces connecting integrated parts as short as possible to minimize parasitic inductance.
- Use wider traces to connect power or ground planes to embedded capacitors in order to reduce resistance. Wider traces also enhance current-carrying capacity and minimize voltage drops.

- For high-speed signals, ensure consistent impedance by maintaining uniform trace width and spacing, referencing the appropriate ground or power plane.
- Design pads to match the size and shape of the component's contact points. This ensures reliable electrical and mechanical connections during fabrication. For example, embedded resistors may require rectangular pads to align with their geometry, while capacitors may need larger pads to accommodate electrode areas.
- For parts that dissipate significant power, utilize thermal relief pads. This reduces the risk of thermal stress during lamination and operation.
- Avoid placing vias in series with integrated components, as they might introduce parasitic inductance that can degrade performance.
- Account for fabrication tolerances in trace and pad designs. Variations in etching or lamination can affect component performance. Specify tolerances in the fab notes to ensure consistency.



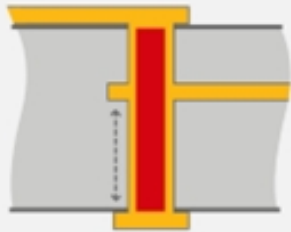
7. How to achieve uniform impedance with embedded passives

Maintaining uniform impedance is critical in high-frequency circuits, especially when embedded passives are a part of signal paths. Even slight variations in impedance might lead to reflections, signal degradation, and timing errors.

7.1 Design tips to ensure consistent impedance with integrated parts

- Modify trace width, thickness, and clearance to match the line impedance. The thicker the copper, the lower the impedance.
- Design all high-speed signal traces with a consistent width and spacing. Avoid tapering or abrupt width changes along the trace.

- Route traces over continuous ground or power planes to provide a uniform return path and avoid impedance discontinuities.
- Avoid stubs and unnecessary via transitions. These can introduce impedance mismatches unless carefully modeled and compensated.



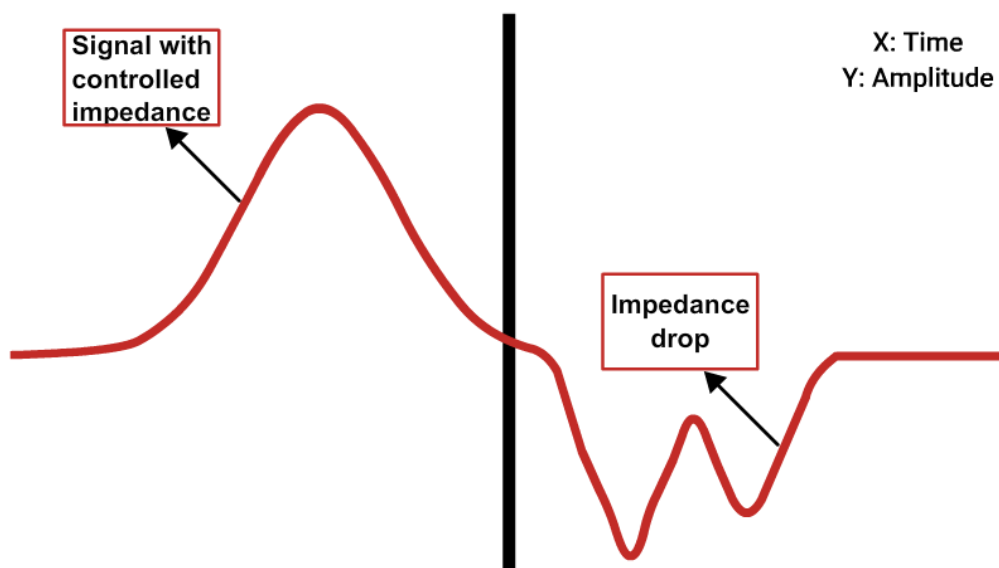
PCB DESIGN TOOL

Maximum Via Stub Length Calculator



TRY TOOL

- When integrating embedded resistors or capacitors, align them with the signal path and avoid sharp turns.
- Before moving to fabrication, simulate your design using signal integrity tools to identify impedance mismatches.
 - 3D field solvers can model the real-world effects of embedded passives and surrounding geometries, including nearby planes and dielectric materials.
 - Time-domain reflectometer (TDR) helps you visualize impedance changes along a trace.



TDR plot detects the locations of impedance discontinuity

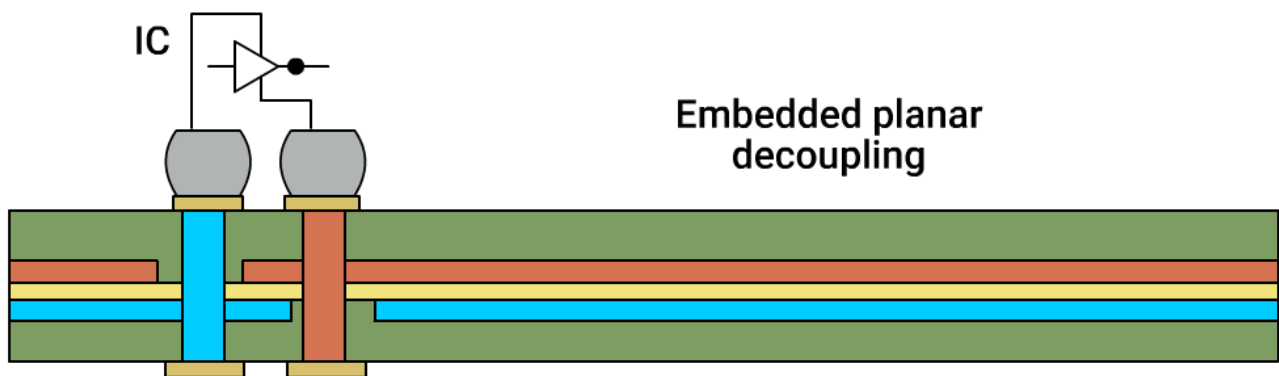
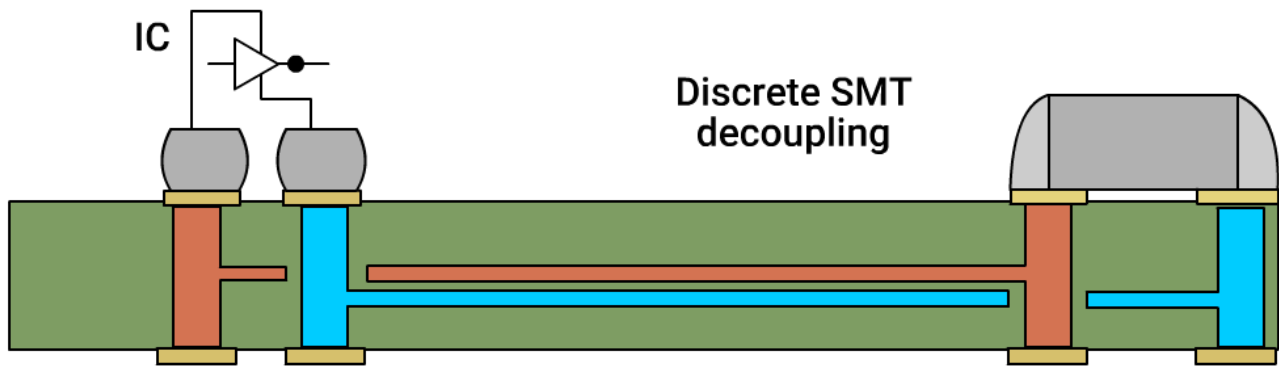
8. Decoupling strategies with embedded capacitors

Embedded capacitors offer minimal parasitic effects, making them ideal for high-frequency decoupling. In this section, you'll learn how to effectively use embedded decoupling capacitors in your layout to suppress high-frequency noise.

8.1 Best practices to place embedded decaps in your layout

Crosstalk and EMI often originate from fluctuations and noise within the power delivery network. Embedded capacitors help suppress these effects by offering a low-impedance path for high-frequency transients. Follow these guidelines when placing decaps:

- Place capacitors directly between power and ground planes to reduce loop inductance. By integrating capacitors as thin dielectric layers between internal power and ground planes (e.g., embedded planar capacitors), you significantly reduce the effective loop inductance.
- Select capacitors with low (ESR) to maximize energy absorption at high frequencies and minimize voltage ripple.
- To maximize decoupling efficiency, locate decaps as close as possible to the power inputs of critical ICs. This reduces the power delivery path impedance.
- Keep the traces or vias connecting embedded capacitors to power pins as short and direct as possible. This minimizes parasitic inductance and improves capacitor effectiveness, particularly at high frequencies.
- Use multiple small-value capacitors distributed across the board to minimize impedance peaks. Power delivery networks have a frequency-dependent impedance profile. Distributed embedded capacitors (1-10 nF range) placed close to ICs help dampen resonance and maintain a flatter impedance curve.
- Position embedded capacitors near sensitive high-speed components (e.g., SerDes, DDR, RF modules) to locally suppress noise and reduce coupling between adjacent traces or power domains.



Discrete SMT and embedded planar capacitor for decoupling



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Power Distribution
Network Analyzer



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8.2 Combining discrete and embedded caps for effective decoupling

No single capacitor type can provide optimal performance across the full frequency spectrum. A hybrid decoupling strategy leverages the strengths of both embedded and discrete capacitors.

Stick to the following strategies to implement a hybrid approach:

- Use embedded capacitance layers for high-frequency decoupling (typically above 100 MHz). At these frequencies, SMT capacitors may not be effective due to parasitic inductance.
- Deploy surface-mount MLCCs (multi-layer ceramic capacitors) with higher capacitance values such as 100 nF, 1 μ F, or 10 μ F, for handling lower-frequency switching transients and bulk energy storage.

Create a model of your power distribution network and simulate it to ensure the combination provides a flat impedance profile across the frequency range.



9. Thermal management techniques for embedded circuits

As PCB designs become more compact, thermal management becomes critical to ensure long-term reliability. In this section, you'll learn how to manage thermal hotspots in embedded designs and apply effective thermal via strategies to ensure reliable heat dissipation.

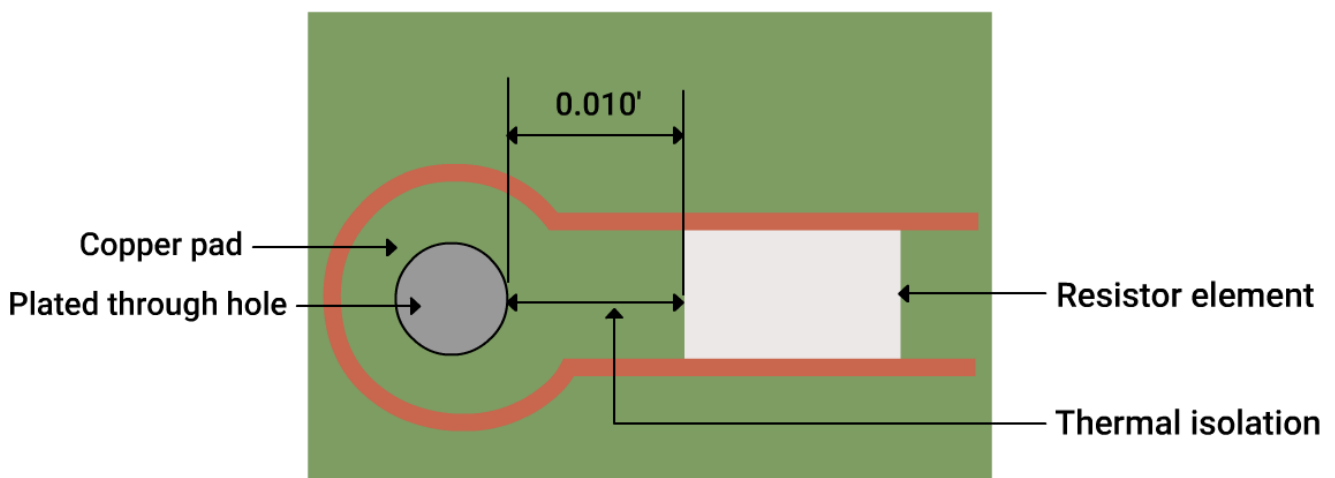
9.1 Identifying thermal hotspots in embedded designs

Embedded components are often located deep within the board stack-up, where airflow is limited and traditional cooling techniques like heatsinks or fans are not so effective. Therefore, accurately managing hotspots is critical.

Here's how you can identify and mitigate hotspots:

- Use thermal simulation tools to model heat generation and flow across the board. Focus on high-power embedded elements like resistors or power ICs to predict where hotspots are likely to form.
- Conduct power density analysis to identify areas with high power dissipation per unit area. These zones are at greater risk of overheating and require targeted thermal strategies.

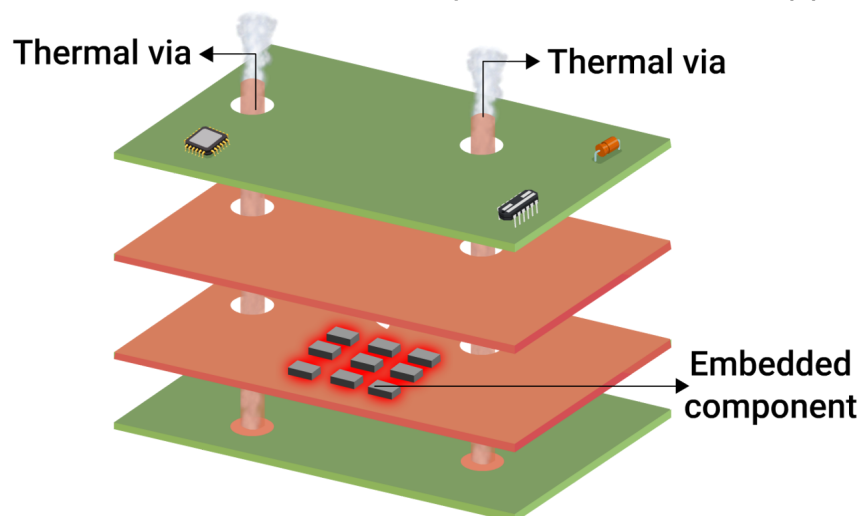
- Monitor temperature gradients across PCB layers using temperature sensors during prototyping. Correlate sensor data with simulation results to validate hotspot locations and assess cooling effectiveness.
- Choose thermally conductive materials such as laminates with thermal conductivity greater than 170 W/m-K to help dissipate heat more efficiently from the inner layers.
- Do not cluster heat-generating embedded components. Distribute them evenly to reduce localized heating and improve overall thermal balance.
- To minimize stress during soldering and prevent resistance shifts from board movement or drilling, maintain proper thermal and mechanical isolation. Keep at least 10 mil clearance between plated through holes and resistor elements, and 5 mil for laser-drilled microvias.



Thermal isolation is necessary to reduce the stress during soldering

9.2 Thermal vias

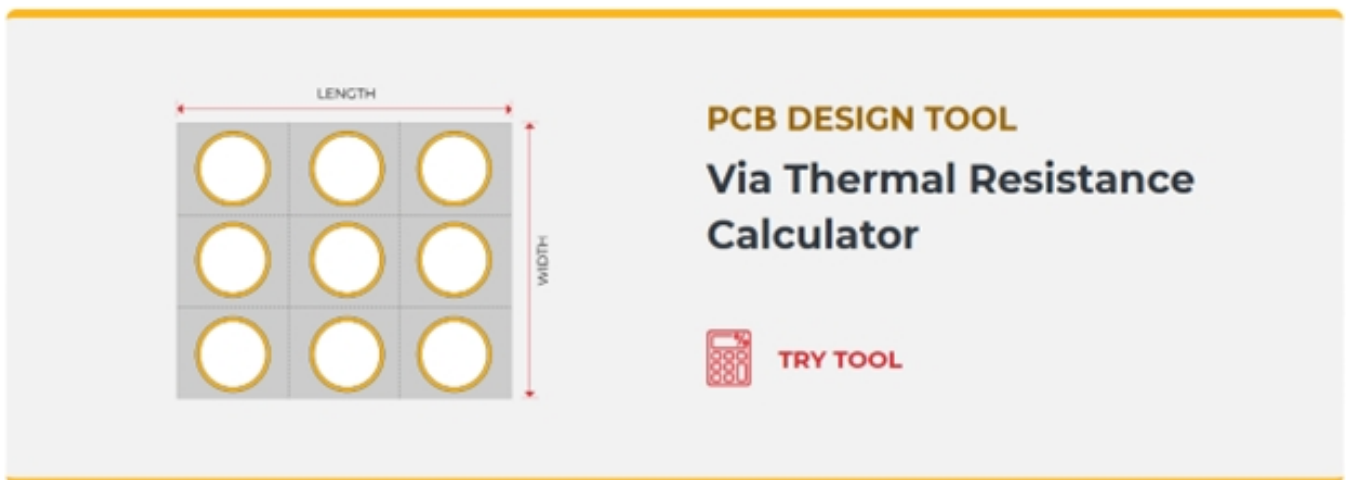
Thermal vias provide vertical heat conduction paths that help draw heat away from embedded parts and distribute it to outer layers, heat sinks, or copper pours.



Position thermal vias beneath embedded components to create efficient heat dissipation paths

9.2.1 Best practices for designing thermal vias

- Place thermal vias beneath or adjacent to embedded components.
- Use via arrays for better heat spreading. Arrays of small-diameter vias (typically 0.3 mm or smaller) can collectively provide a high thermal transfer rate without significantly affecting the board's electrical behavior.
- For enhanced thermal performance, consider:
 - Solid-filled vias (e.g., copper-filled) as they offer maximum conductivity.
 - Conductive epoxy-filled vias when you need planarity or additional structural integrity.
 - Via-in-pad technology for direct thermal contact with embedded dies or copper planes.
- Ensure a sufficient number of vias while maintaining electrical isolation. As a rule of thumb, maintain a via pitch of 1 mm or less in high-heat zones.
- Thermal vias must not obstruct critical signal paths or interfere with embedded passive components. Collaborate with your manufacturer to balance thermal and electrical design requirements.

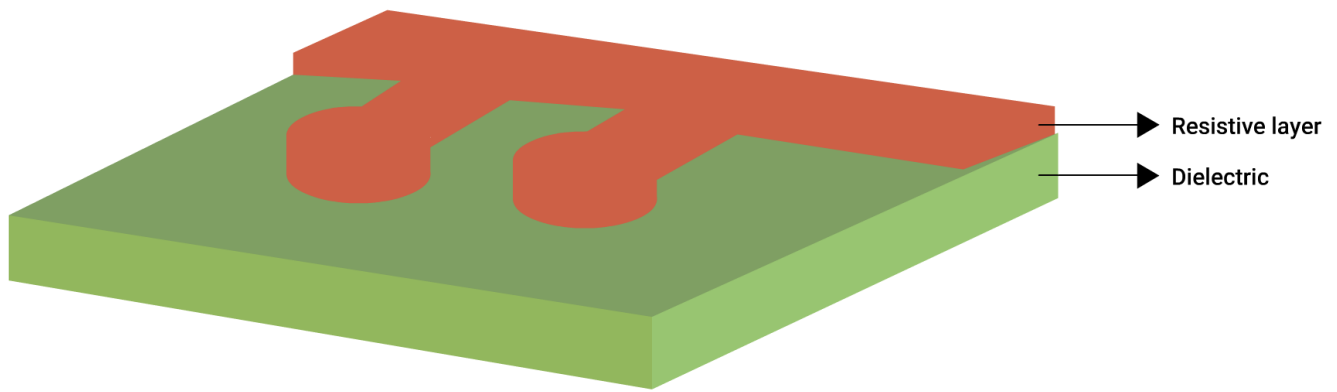


10. How are embedded passives fabricated

10.1 Additive and subtractive fabrication methods

Additive method: It involves directly depositing or building up resistive, capacitive, or inductive materials on the substrate.

Additive methods are generally cost-effective for high-volume production due to reduced material waste and simplified processes. It offers flexibility in material choice but may have limitations in achieving ultra-fine features and very tight tolerances compared to some subtractive methods.



Inclusion of a resistive layer on a PCB

Subtractive methods: This process etches away excess material to form resistors or capacitors (e.g., thin-film resistors). The method offers higher precision but is more expensive.

Subtractive methods, particularly those involving photolithography, offer higher precision, tighter tolerances, and finer features. However, they typically involve more processing steps, consume more material, and are generally more expensive. Thin-film resistors and capacitors are commonly fabricated using these methods.

10.2 Screen printing and laser trimming

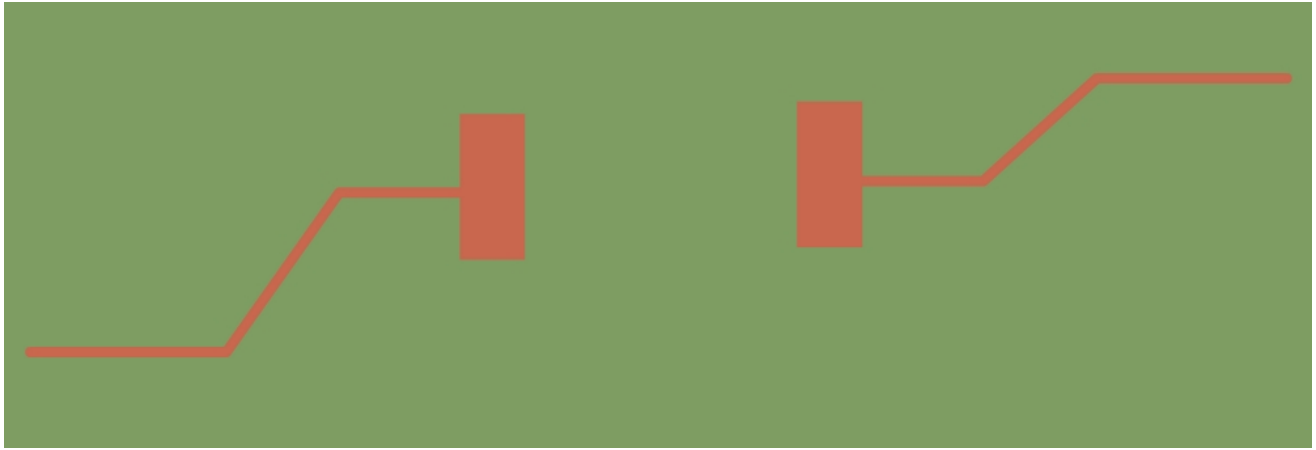
Screen printing: Used to fabricate thick-film resistors and some capacitors, where resistive or dielectric paste is applied through a stencil. It's cost-effective but less precise.

Laser trimming: The process adjusts the resistance or capacitance by selectively removing material with a laser. This improves tolerances but increases fabrication time and cost.

OhmegaPly and TCR are commonly used resistive materials used to fabricate embedded resistors. In the next section, you will learn how passives are created using these materials.

10.3 Step-by-step fabrication process of OhmegaPly

OhmegaPly processing is a subtractive manufacturing process. The fabrication begins with a laminate that has a thin resistance film under the copper. In the initial production stage, the pad and trace layers are merged with the resistor body using Gerber, extended Gerber, or ODB++. Below is an illustration of a trace connected to an embedded resistor.

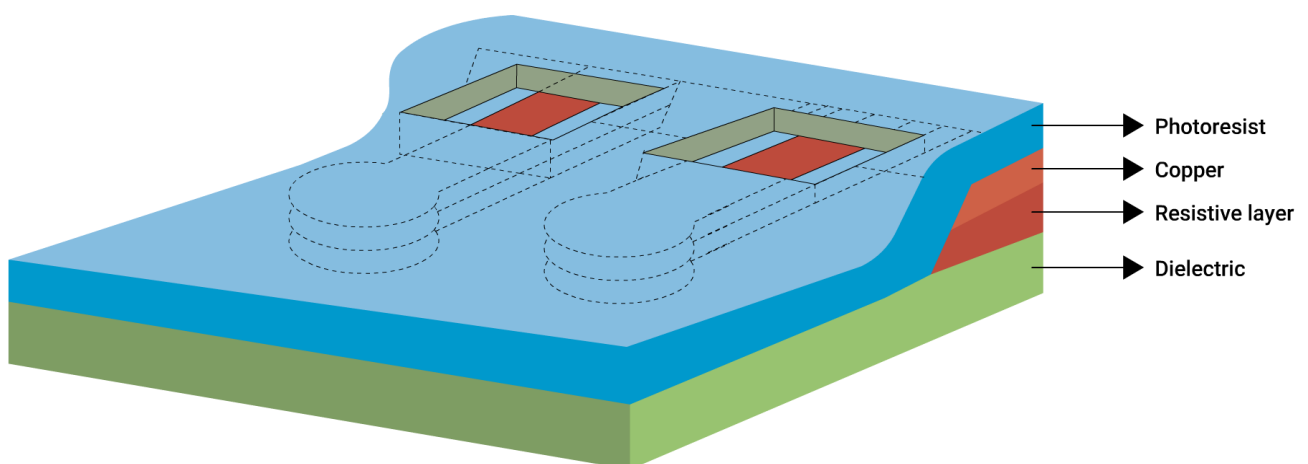


Trace connected to an embedded resistor

Given below is the list of sequential steps and their representations for processing OhmegaPly.

Step 1: Application of photoresist

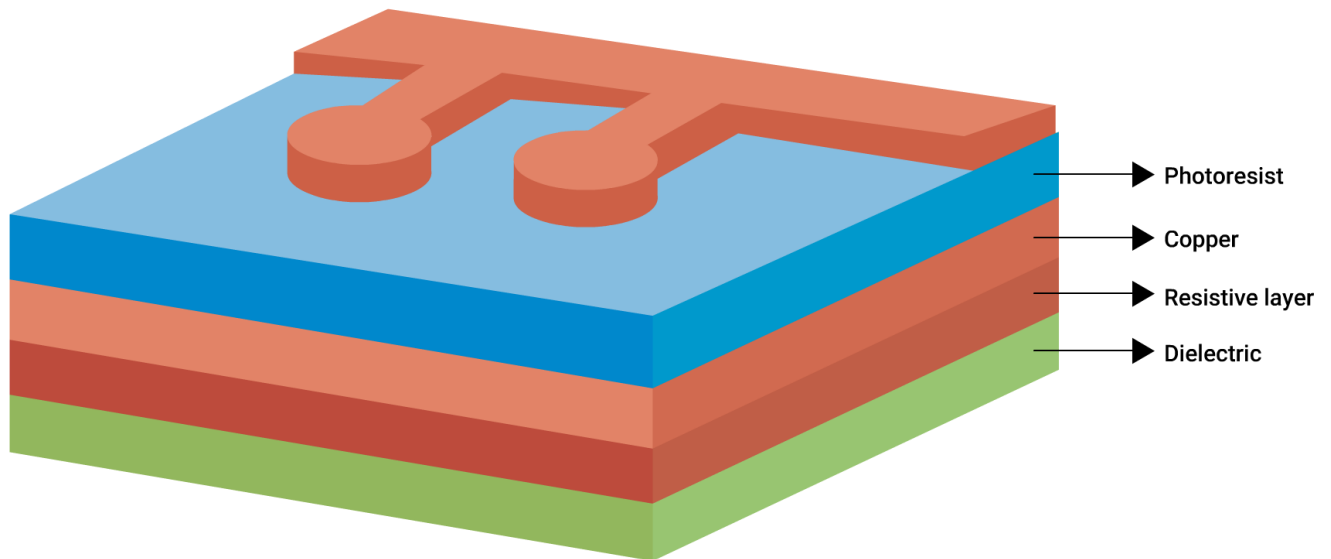
This step is similar to multilayer board manufacturing. In this step, a photoresist (a layer of the photosensitive film) is applied to the OhmegaPly laminate. It polymerizes upon exposure to UV light.



A photoresist layer is applied to the OhmegaPly laminate

Step 2: Printing and developing

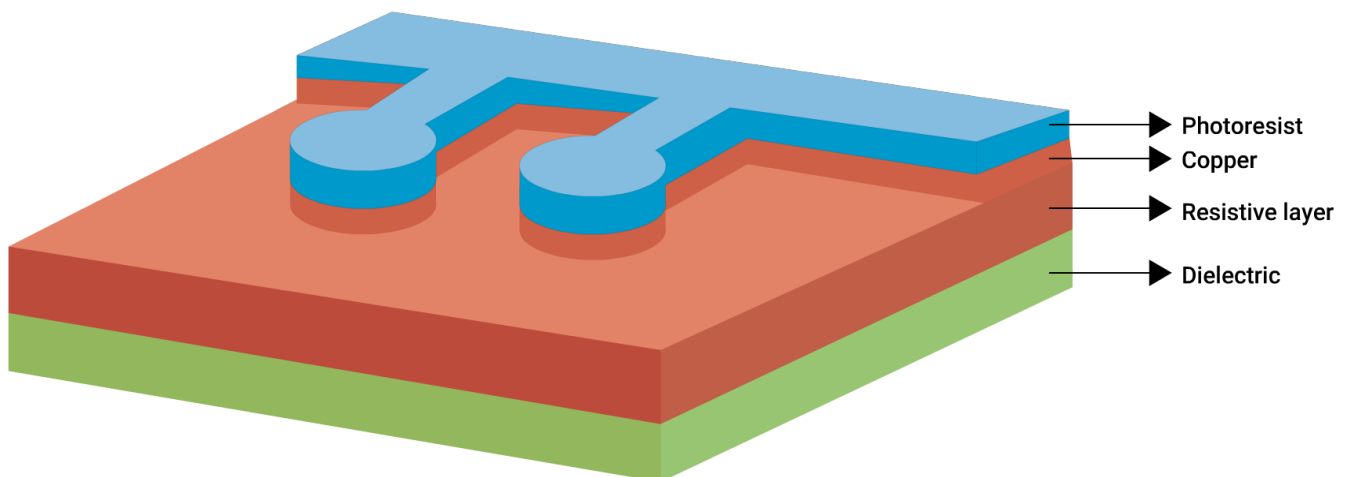
This phase involves printing and developing a composite image. The panel is placed under a computer-controlled laser. The system scans the board surface and generates a digital image that matches the specifications from a pre-loaded CAD/CAM file.



Printing and developing resistive layers using photoresist

Step 3: Copper etching

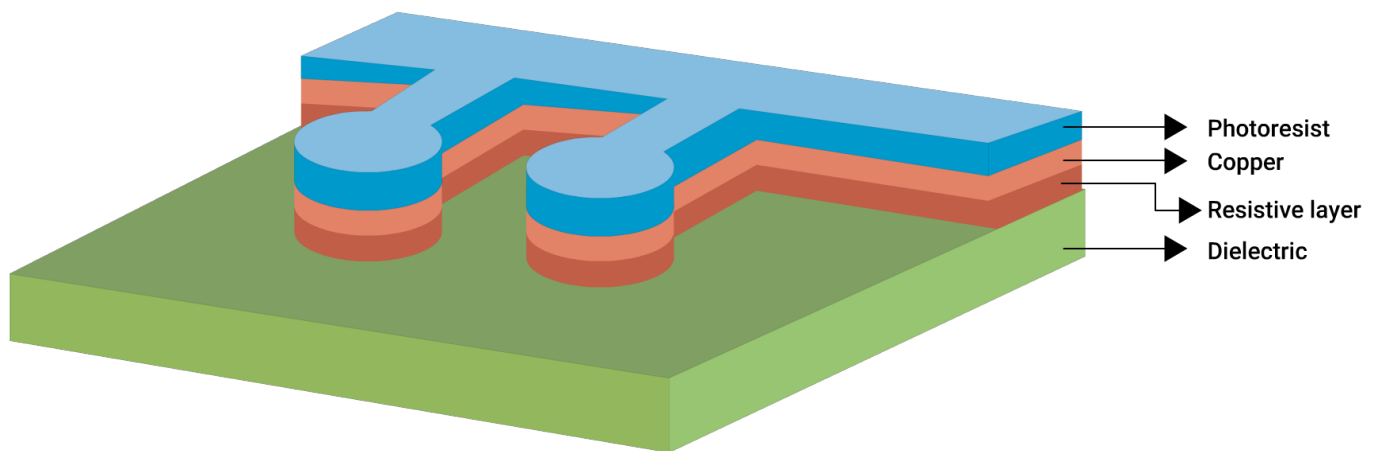
In this stage, unwanted copper is etched away using a conventional etchant. This creates the desired copper circuit pattern.



Etching removes the unwanted copper

Step 4: Resist etching

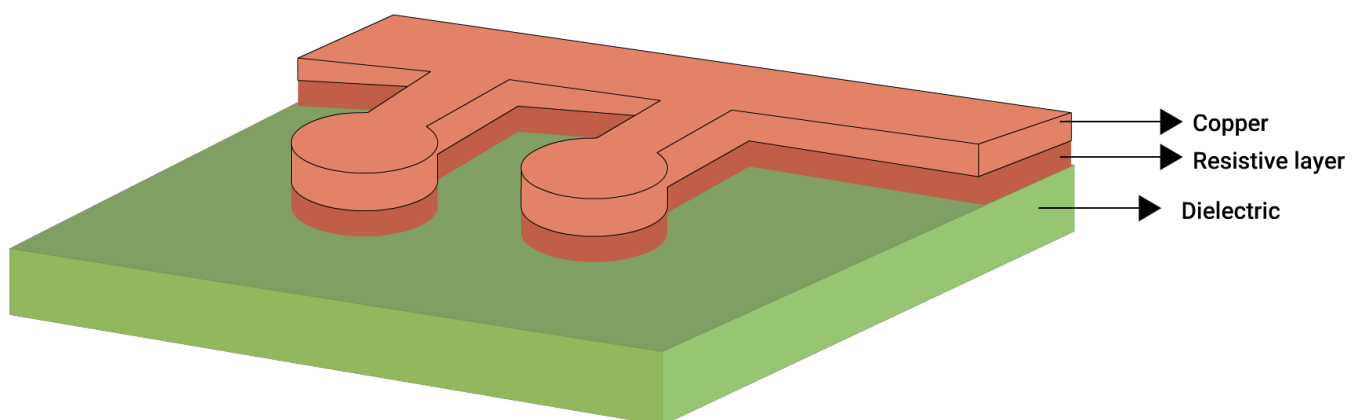
After copper etching, the unwanted resistive material is etched away using a copper sulfate (CuSO_4) solution. This is often called as the second etch process.



Unwanted resistive material is etched using CuSO_4 solution

Step 5: Photoresist stripping

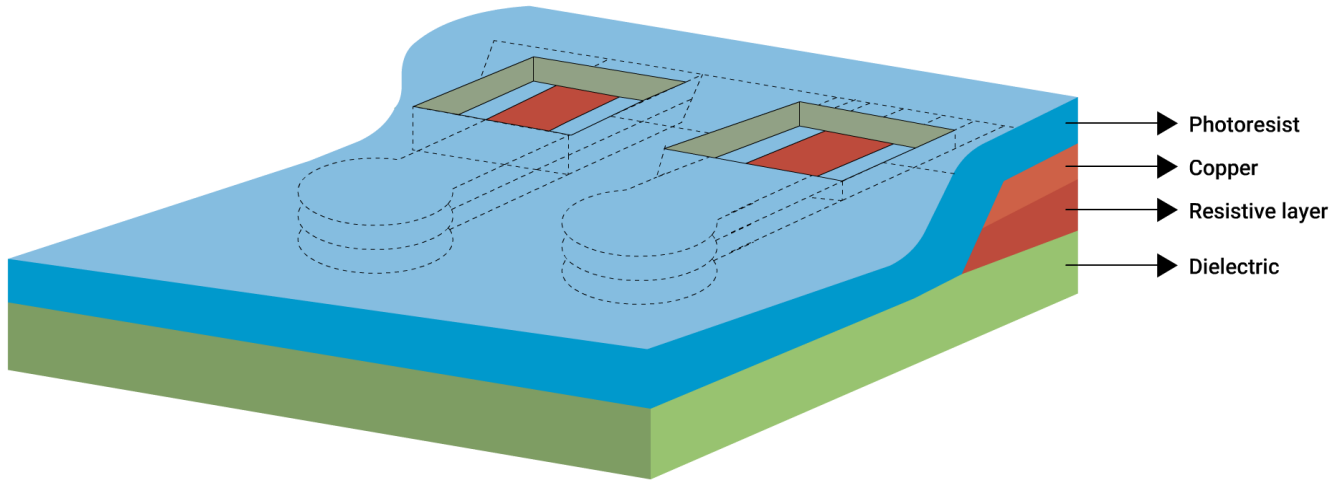
In this step, photoresist is stripped from the copper surface using caustic agents dissolved in water. This is followed by a high-pressure rinse to completely clean the surface.



Remaining photoresist is removed from the copper

Step 6: Photoresist application

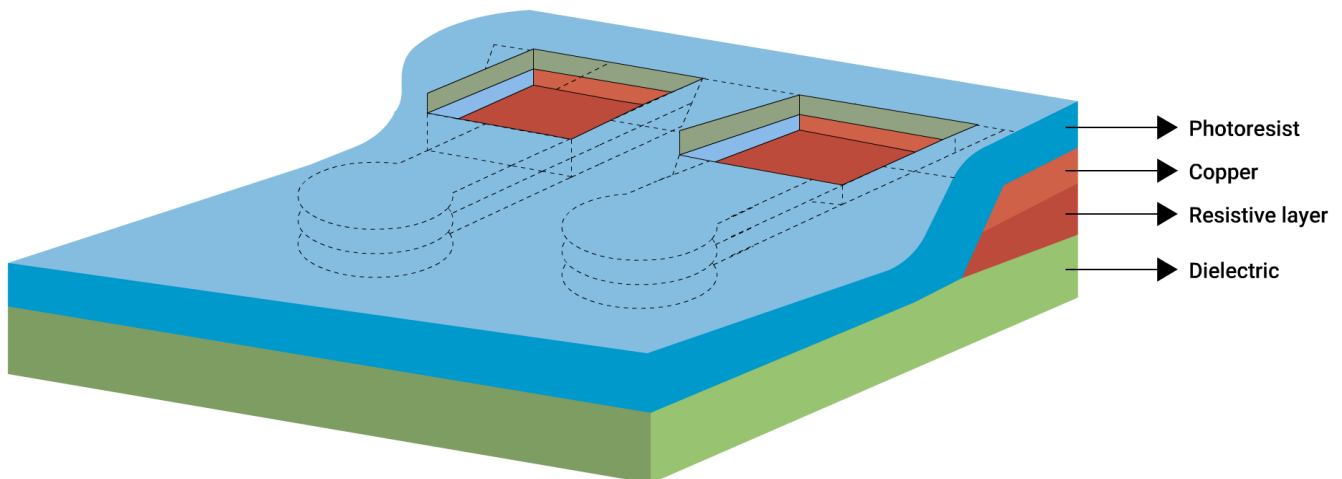
In this step, the photoresist is applied again. This is followed by printing and developing the conductor-protect image. This is also called the second printing process.



The photoresist is applied again

Step 7: Etching

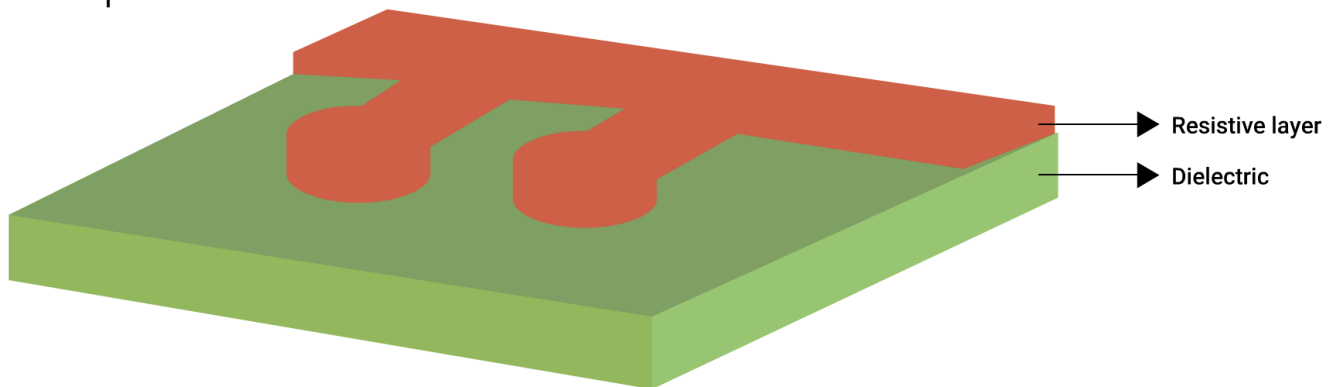
Excess copper is etched away from the resistor design area using a selective alkaline etchant.



The excess copper is etched away from the designed resistor area

Step 8: Photoresist stripping

This is the final step, the photoresist is stripped to reveal the completed embedded resistor pattern.



Final pattern of an embedded resistor in a PCB

10.4 Fabrication steps for TCR embedded resistors

TCR processing follows a similar flow to OhmegaPly but includes key differences, particularly in material handling and etching steps.

Step 1: Application of photoresist

Like OhmegaPly, TCR-based embedded resistor fabrication follows conventional multilayer PCB processing. The resistive material is integrated into the layer stack-up as part of the inner layers. The foil is laminated to form an inner layer core.

Step 2: Printing and developing

The photoresist is exposed and developed using a laser system based on a CAD/CAM image. This defines the pattern for copper and resistive elements.

Step 3: Copper etching

Unwanted copper is etched using cupric chloride, just like in standard processes. This step also begins to remove NiCr, since it's etched alongside the copper.

Unlike OhmegaPly, TCR does not need a second etch step (resistive etching with CuSO_4). The NiCr alloy is removed along with copper using the same etchants, simplifying the process.

Step 4: Photoresist stripping

Remaining photoresist is removed using caustic solutions and rinsed with high-pressure water.

Step 5: Second photoresist application

If needed, a second photoresist layer is applied for defining the conductor-protect image, just as in OhmegaPly.

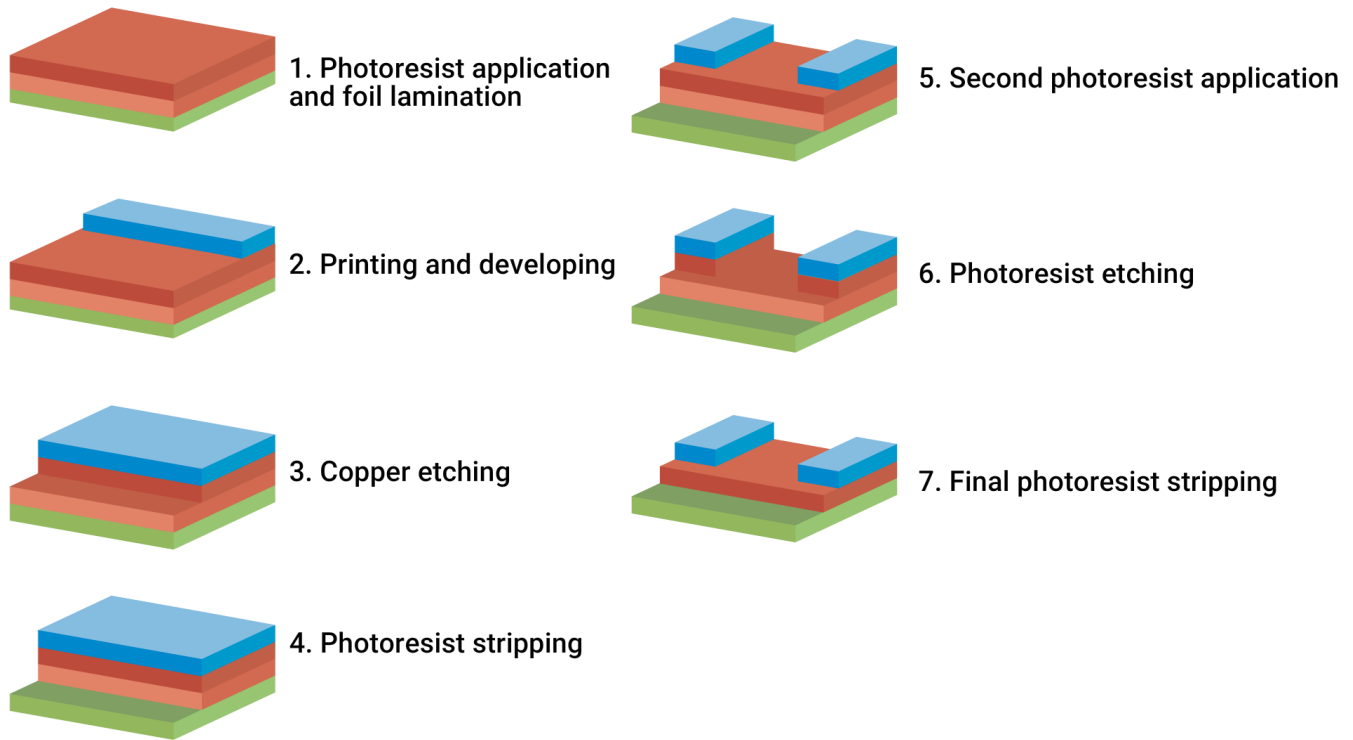
Step 6: Etching

Excess copper over the resistor pattern is etched away using a selective alkaline etchant, ensuring clean resistor definition.

Step 7: Final photoresist stripping

The final photoresist is stripped, revealing the finished embedded resistor pattern.

The steps are summarized in the image below:



Fabrication steps for TCR embedded resistors

11. Communicating embedded passive requirements to your manufacturer

Effective communication with your manufacturer is critical to ensure the successful production of circuit boards. Fab notes are a vital part of PCB design package. They provide clear instructions to your CM.

11.1 What should you include in your fab notes?

You should clearly define technical requirements in your fab notes to avoid ambiguity and ensure the board is fabricated and assembled correctly.

In your fab notes, include:

- Component type (resistors, capacitors, inductors) and the layer number on which the part should be fabricated.
- Detailed stack-up, indicating the material types and thicknesses of each layer.
- Precise XY locations of each embedded part, along with key specifications such as value, power rating, and physical dimensions.
- Exact depth of each cavity where a component will be embedded. The depth must correspond to the component thickness, with additional margin for adhesive layers or potting material, if used.
- Geometry of the cavities (rectangular, circular, etc.).
- Complete datasheets for each embedded part. These should include electrical specs (e.g., resistance, capacitance, inductance), mechanical dimensions, thermal limits, and any special assembly guidelines (e.g., maximum reflow temperature).
- Tolerances and special requirements. We'll take a closer look at these parameters in the next section.

Here's an example fab note that clearly calls for an embedded resistor:

1. FABRICATE PER LATEST REVISION OF IPC-6012, CLASS 3 SPECIFICATION.
2. FABRICATE PCB PER SUPPLIED CAD DATA. USE FILM TO CHECK.
3. FINISHED BOARD THICKNESS 62 MIL + 10% MEASURED OVER AREA WITH SOLDERMASK.
4. ALL FINISHED BOARDS TO BE 100% ELECTRICALLY TESTED PER SUPPLIED
5. IPC NETLIST. DO NOT EXTRACT NETLIST FROM GERBERS.
6. ALL PLATED THRU HOLES SHALL NOT BE OVER DRILLED BY MORE THAN
7. 0.006 INCH OVER NOMINAL FINISHED HOLE SIZE.
8. EMBEDDED RESISTOR LOCATED ON LAYER L3 BETWEEN COORDINATES X50.0, Y25.0 AND X60.0, Y25.0.
9. APPLY LIQUID PHOTO-IMAGEABLE SOLDERMASK OVER BARE COPPER ON BOTH SIDES, COLOR GREEN|
10. ALL COPPER WEIGHTS REFER TO FINISHED WEIGHT.
11. ALL EXPOSED COPPER FEATURES ARE TO BE FINISHED WITH ENIG.
12. SILKSCREEN TO BE WHITE, NON-CONDUCTIVE, EPOXY INK OR EQUIVALENT.
13. SILKSCREEN INK IS NOT PERMITTED ON ANY SOLDERABLE SURFACES.
14. SILKSCREEN ALLOWED TO ENCROACH ON TENTED VIAS.

Specify the layer and location where an embedded resistor needs to be added in your PCB fab notes

11.1.1 Tolerance specifications

Tolerance defines the acceptable limits for various PCB features.

For embedded components, specify the acceptable tolerances of their electrical properties.

For example,

- Resistors: $\pm 1\%$ tolerance for precision applications or $\pm 5\%$ for general-purpose designs.
- Capacitors: $\pm 10\%$ tolerance for standard applications or $\pm 2\%$ for high-frequency circuits.
- Exact dimensions and tolerances for cavity size.
 - Cavity width and length: $2.00 \text{ mm} \pm 0.05 \text{ mm}$.
 - Cavity depth: $0.15 \text{ mm} \pm 0.02 \text{ mm}$.
- Minimum spacing required between the cavity edges and the component body to accommodate manufacturing tolerances and ease of assembly.

11.1.2 Special requirements

When specifying special features in fabrication notes, the goal is to ensure accurate placement and mechanical integrity of embedded parts.

Here are some important special features that should be documented in the fab notes:

Cavity details:

- **Chamfers:** Angled cavity edges to ease component insertion and reduce mechanical stress.
- **Radii (rounded corners):** Prevent stress concentration and help with milling tool paths.
- **Depth control:** Controlled depth milling for accurate vertical alignment of components.

Encapsulation and potting requirements:

- Type of encapsulant or resin (e.g., thermally conductive, low-viscosity).
- Curing temperature and profile.
- Required clearance for potting material flow.

Adhesive or bonding layer:

- Type of adhesive material (e.g., thermally conductive epoxy).
- Thickness and application method (e.g., screen printing or dispensing).

Special handling instructions for:

- ESD-sensitive components.
- Fragile components: specify limited pressure during lamination or pressing.
- Moisture-sensitive parts.



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Other resources on embedded components

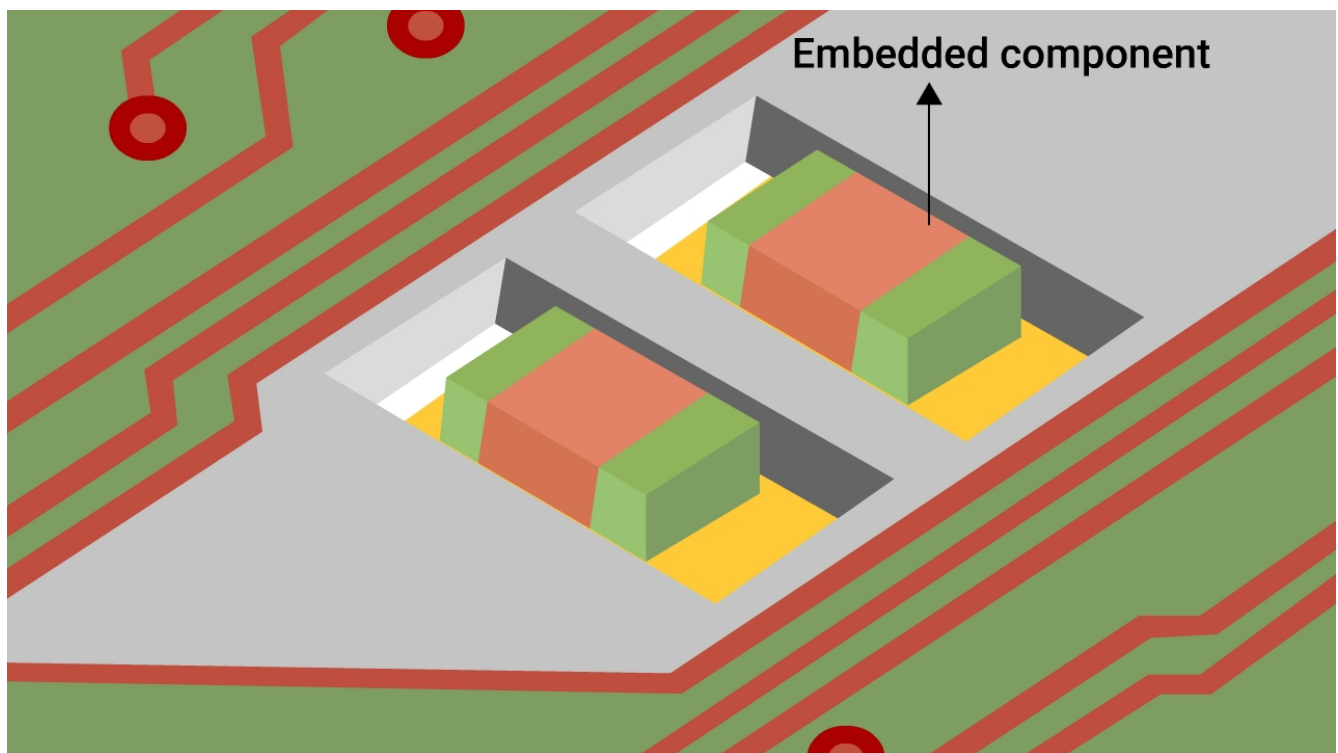


Articles

- [How to Design a PCB with Embedded Components](#)
- [Decoupling Capacitor Placement Guidelines Every PCB Designer Should Know](#)
- [Component Placement Guidelines for PCB Design and Assembly](#)
- [PCB Stack-up: Plan, Design, Manufacture and Repeat](#)

Webinars by industry experts

- [Improving PCB Performance with Embedded Resistors and Capacitors](#)



An illustration of embedded passives integrated into a PCB substrate

About Sierra Circuits



Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Talk to our experts

Sierra Circuits helps PCB designers plan it right! Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view. Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Why Sierra Circuits?

Customer-oriented approach: We collaborate closely with you and tailor solutions to meet your unique design requirements.

Rapid high-tech innovation: We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

Integrated systems: Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages at the beginning of all of our partnerships.

Focus ready: We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits’ PCB manufacturing capabilities overview

Rigid PCBs

PCB highlights	Fabrication highlights	Quality management systems
Up to 30 layers	Routed arrays	AS9100D
Board thickness: .005" - .250"	Edge plating	ISO 9001:2015
Max panel size: 21" x 29"	Bevels	ISO 13485:2016
Min trace and space: .002"	Heat sinks	
Solder mask feature tolerance: .001"		
Blind and buried vias		

HDI and microelectronics

High tech in small form	Features	Certified for your industry
Special stack-up help / NPI review	Down to 1.5-mil trace space, 2-mil holes	Mil-spec
Up to 30 layers	Blind vias, buried vias, and other microvia techniques	Automotive
Files evaluated by engineers	Up to 5 sequential lamination cycles	Aerospace and defense
Quote sent by email within a few hours	Laser direct imaging	Medical devices
	via-in-pad technology	

Flex and rigid-flex PCBs

Surface finishes	Standard flex materials	Fabrication highlights	PCB classification
HASL	0.5 mil to 5 mil thick polyimide material	Routed array	IPC 6012, Class 1, 2, and 3
Lead-free HASL		V score	
OSP (Shikoku F2 and Entek)	1 mil to 5 mil thick copper clad base material	Edge plating	ISO:9001:2015
ENIG	Flame retardant laminate	Edge castellation	ISO 13485: 2015
Immersion silver		Countersink	
Immersion white tin	UL and RoHS compliant	Bevel	
Tin nickel		Milling	
Selective gold			

Aerospace and defense PCBs

Capabilities	Documents we provide	Ideal for
Layer count: Up to 30	Certificate of conformance	DOD product designers developing systems for mission critical functions
Minimum board thickness:10 mil	Material specifications	
Thickness tolerance: 10%	Reflow profile copy (included with first article)	Contract manufacturers who need fast, predictable lead times
Minimum trace and space width: 3 mil	Photo requirements	
Minimum drill-to-copper clearance: 6 mil	First article inspection report	Purchasing teams who require pricing and supply-chain transparency
Maximum panel size: 21" x 29"	Record of calibrated tools used during manufacturing	
Controlled impedance	FPT, AOI, ionic cleanliness report	Aerospace engineers looking to shore-up their supply chain troubles
Stacked vias		

RF and microwave PCBs

Capabilities	Certifications
Operating frequency: 10 MHz to 30 GHz	PC Manufacturers Qualification Profile (MQP)
Sodium etch hole treatment process	IPC-A-600
Full-body gold plating up to 120u"	IPC-6012 standards
Soft gold without nickel or ENEPIG surface finishes for enhanced board protection	ISO 9001:2015
	NADCAP
	ITAR registered
Design assistance	RoHS/REACH standards

Sierra Circuits’ PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

Full turnkey PCB	Partially consigned assembly	Consigned assembly	Assembly capabilities
Design review	Some parts provided by the designer	Components are provided by the PCB designer	BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components
Bare board fabrication	We will source the rest	Sierra Circuits will fully assemble your PCBs and ship them to you.	DFA
Component procurement	Supply chain resilient	Quick turn times	RoHS, leaded, indium, clean and no clean chemistry
Testing	Quick turnaround time	Fully transparent pricing	Component sizes: 0201, 01005, 08004
Quick turnaround time			Paste in-hole

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

Sierra Circuits' PCB design services

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of $\pm 5\%$
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

Advanced PCB design tools

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

Sierra Circuits PCB certification and registration

IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace, and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D – military and aerospace
- ISO 13485:2016 certificate – medical devices

Mil-Spec certified and DLA approved

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid flex PCBs
- Rigid PCBs

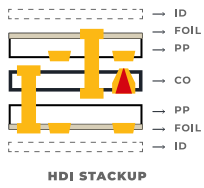
Exceptional material compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

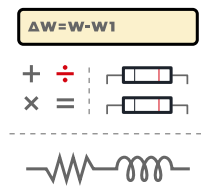
Certified minority supplier by NMSDC

- Minority business certification

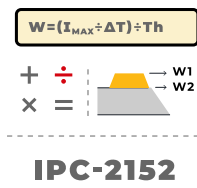
The Designer's Toolkit



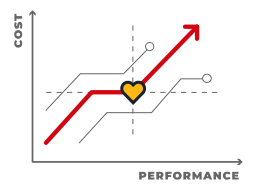
Stackup Designer



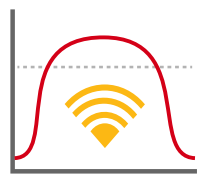
Impedance Calculator



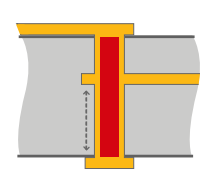
IPC-2152
Trace Width,
Current Capacity and
Temperature Rise
Calculator



Material Selector



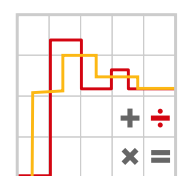
Bandwidth, Rise Time
and Critical Length
Calculator



Maximum Via
Stub Length
Calculator



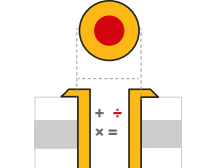
Signal Layer Estimator



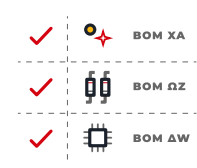
Transmission Line
Reflection Calculator



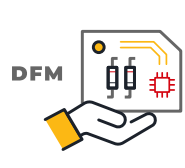
RLC Resonant
Frequency &
Impedance Calculator



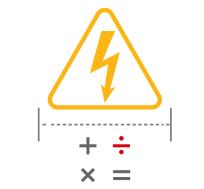
Via Current
Capacity and
Temperature Rise
Calculator



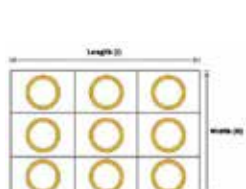
BOM Checker



Better DFM



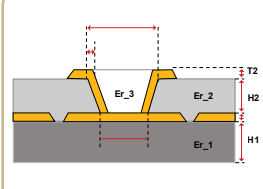
Conductor Spacing
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Via Thermal
Resistance Calculator



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Network Analyzer



Via Impedance
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Try Tools

CONTACTS

Amit Bahl

Chief Revenue Officer
amit@protoexpress.com

Steven Carney

R&D Manager
stevenc@protoexpress.com

Lucy Iantosca

Director of Marketing
lucyi@protoexpress.com

Made in America 

Bare board manufacturing

1108 West Evelyn Avenue Sunnyvale, CA

PCB assembly

1980 Lundy Avenue San Jose, CA



www.protoexpress.com